

PharmaSignal AI

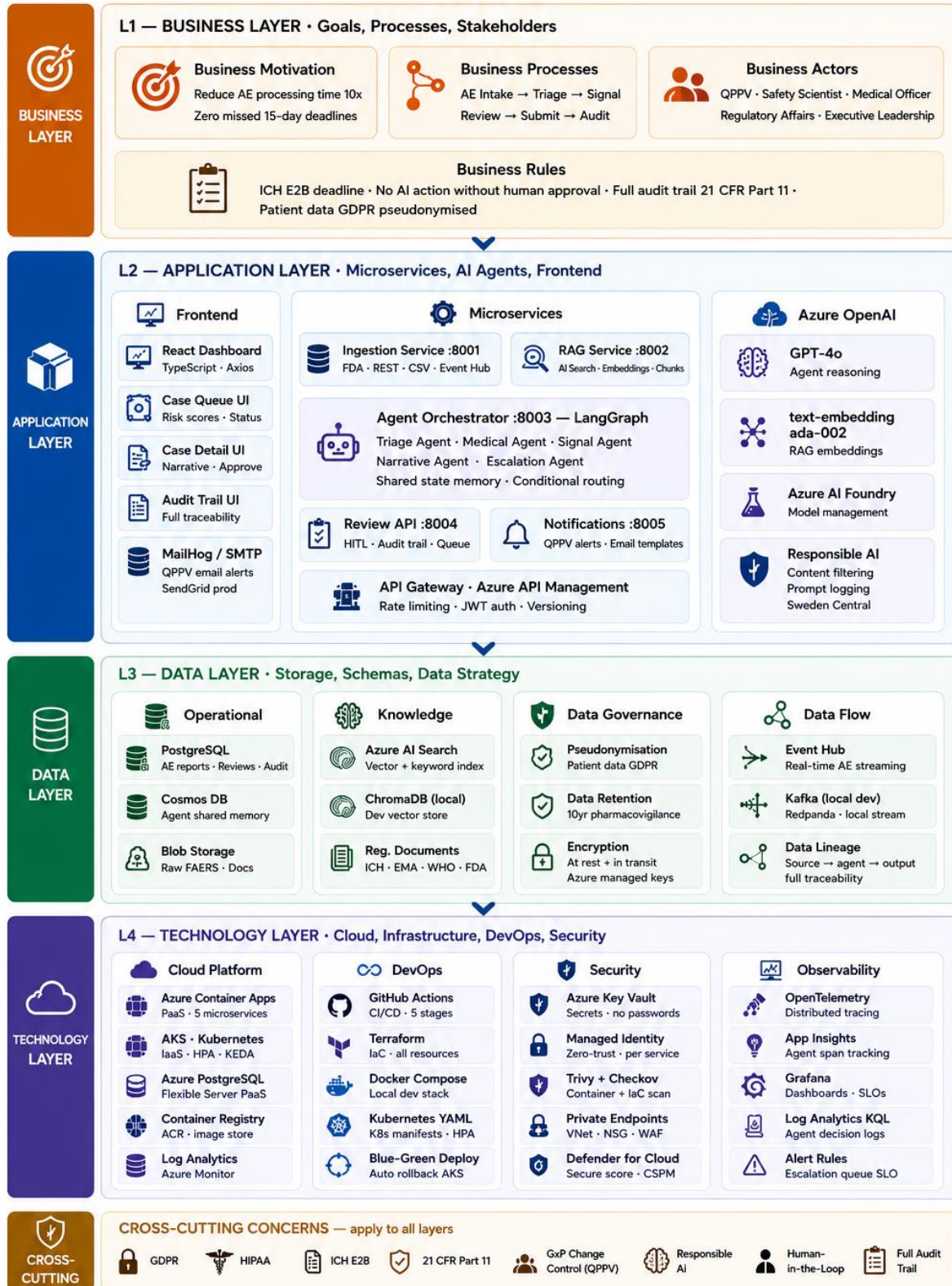
Enterprise Multi-Agent Pharmacovigilance Intelligence Platform

Architecture Diagrams

PharmaSignal AI — Enterprise Architecture

Multi-Agent Pharmacovigilance Intelligence Platform

TOGAF Layered View



TOGAF Layer Mapping

L1 Business Layer

L2 Application Layer

L3 Data Layer

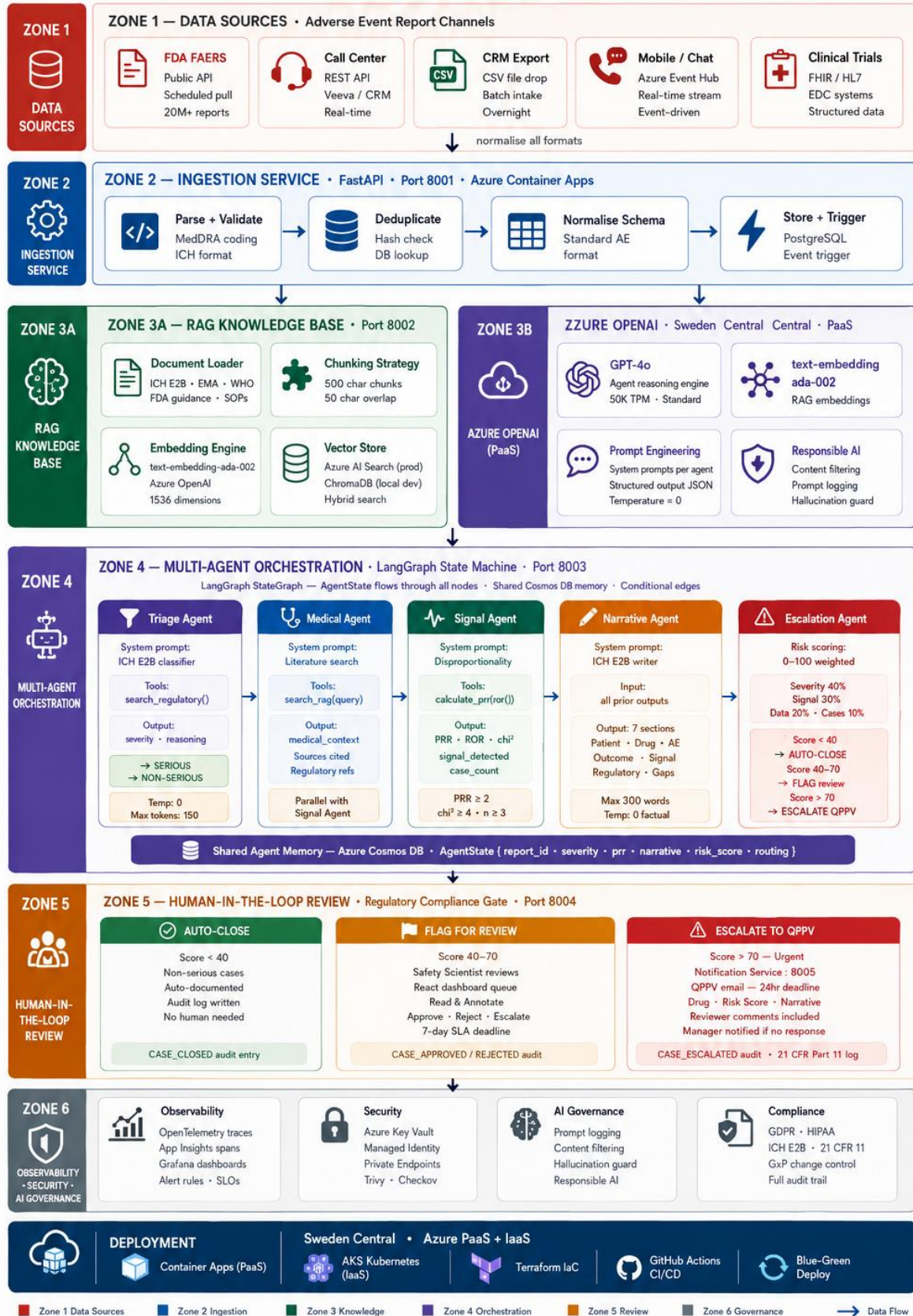
L4 Technology Layer

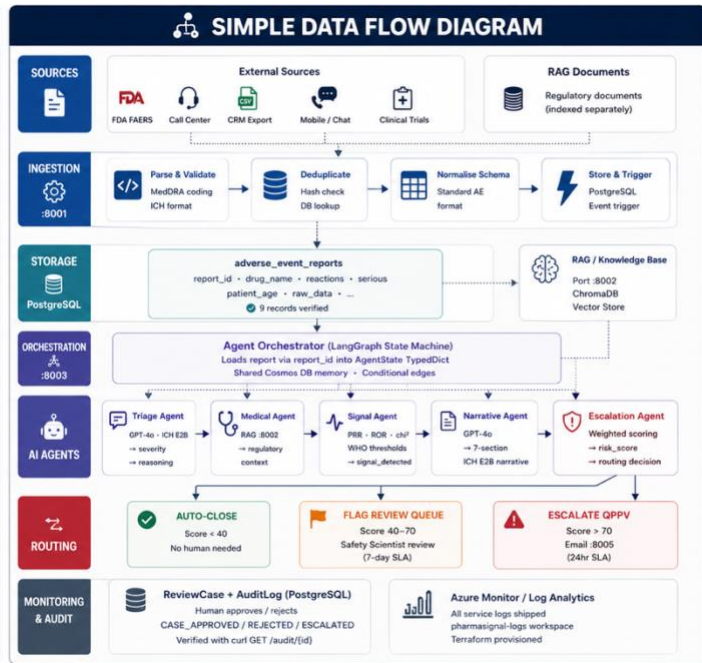
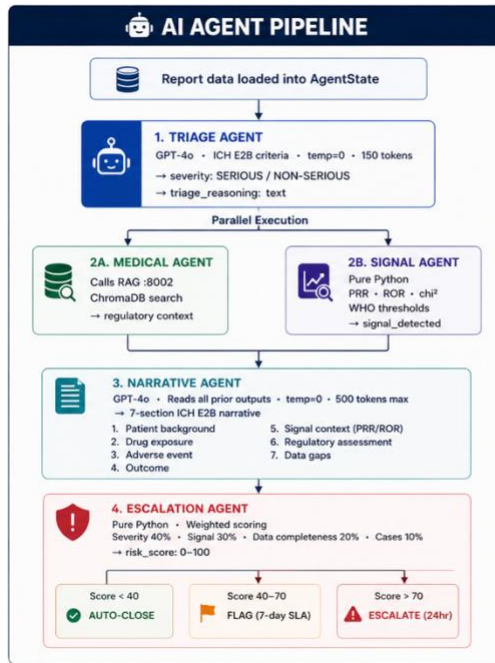
Layer interaction / flow

PharmaSignal AI – AI Solution Architecture

Multi-Agent Pharmacovigilance Intelligence Platform

Azure OpenAI • LangGraph • RAG





PharmaSignal AI — Security & Compliance Architecture

Zero-Trust • Defence in Depth • Regulatory Compliance • Responsible AI



DEFENCE IN DEPTH • ZERO-TRUST ARCHITECTURE



Sweden Central



Azure PaaS



Zero Trust



Least Privilege



Always Verify



No Implicit Trust

PharmaSignal AI — Full Enterprise Governance Architecture

Production Vision • What a Complete Enterprise System Would Have



L1 — AI GOVERNANCE



Human-in-Loop

- No auto-submission ever
- Risk-based routing 0–100
- QPPV escalation chain
- Override capability
- Explainable AI output



Model Governance

- Model version registry
- Prompt versioning
- A/B testing framework
- Drift monitoring
- Performance tracking



Responsible AI

- Content filtering
- Bias detection
- Fairness monitoring
- Prompt injection guard
- Hallucination detection



AI Risk Register

- Risk identification
- Mitigation controls
- Residual risk scoring
- Quarterly review
- Board reporting



L2 — REGULATORY COMPLIANCE FRAMEWORK



GDPR

- Art 25 privacy design
- Pseudonymisation
- Right to erasure
- Data portability
- 72hr breach notify
- DPA agreements
- SCCs transfers



HIPAA

- PHI protection
- No PHI in logs
- Access controls
- Audit controls
- Transmission security
- BAA with Azure
- Breach notification



ICH E2B

- ICSR format R3
- 7 narrative sections
- MedDRA coding
- 15-day deadline
- 7-day expedited
- E2B XML submission
- Narrative quality



21 CFR Pt11

- Electronic records
- Electronic signatures
- Audit trail
- System validation
- IQ OQ PQ
- Access controls
- Computer systems



GxP

- Change control
- 2-person approval
- SOP compliance
- CAPA process
- Validation protocol
- Risk assessment
- Deviation handling



ISO 27001

- Info security mgmt
- Risk management
- Asset management
- Incident response
- Business continuity
- Supplier security
- Annual audit



L3 — AUDIT AND TRACEABILITY



Immutable Audit Log

- Every action timestamped
- Actor • Action • Details
- Cannot be modified
- PostgreSQL AuditLog
- Long-term retention



Agent Decision Trail

- All 5 agent outputs stored
- Reasoning logged
- Risk score history
- Full analysis JSON
- Cosmos DB backed



Change Management

- Git commit history
- PR approvals logged
- Deployment records
- Rollback capability
- Azure Monitor logs



SIEM + Alerting

- Azure Sentinel
- Threat detection rules
- Anomaly detection
- Incident playbooks
- 24hr response SLA



L4 — HUMAN REVIEW GOVERNANCE



Review Workflow

- PENDING queue
- Priority by risk score
- 7-day SLA for FLAG
- 24hr SLA for ESCALATE
- Deadline tracking



Approval Hierarchy

- Safety Scientist L1
- Medical Officer L2
- QPPV L3 final
- Executive override
- RBAC enforced



Conflict of Interest

- Reviewer cannot approve own submitted cases
- Four-eyes principle
- Segregation of duties
- Enforced in system



Notification Chain

- Auto email on escalation
- QPPV + reviewer notes
- Manager if no response
- SMS for critical cases
- SendGrid production



L5 — DATA GOVERNANCE



Data Classification

- PII — patient data
- PHI — health records
- Confidential — reports
- Internal — analysis
- Public — guidelines



Data Lineage

- Source channel tracked
- Transformation logged
- Agent processing trail
- Output provenance
- Full traceability



Retention Policy

- PV data 10yr minimum
- Clinical trial 25yr
- Audit logs permanent
- Auto-delete policy
- Azure Policy enforced



Data Quality

- Completeness checks
- MedDRA validation
- Duplicate detection
- Data gap flagging
- Quality score



L6 — OPERATIONAL GOVERNANCE



SLOs Defined

- 99.5% uptime target
- Serious AE < 4hr process
- Queue < 10 threshold
- Alert rules defined
- Azure Monitor



Incident Response

- Runbook documented
- Severity levels P1–P4
- On-call rotation
- QPPV notification
- Post-mortem process



Business Continuity

- RTO 4 hours
- RPO 1 hour
- DB backup daily
- Geo-redundant storage
- DR tested quarterly



Vendor Governance

- Azure BAA signed
- OpenAI DPA signed
- Annual vendor review
- Sub-processor list
- Supply chain risk

6-LAYER GOVERNANCE FRAMEWORK



L1

AI Governance

L2

Regulatory Compliance

L3

Audit & Traceability

L4

Human Review Governance

L5

Data Governance

L6

Operational Governance

Full enterprise production vision • Associate Director reference architecture